MILITARY SPECIFICATION

CABLE ASSEMBLY, SPECIAL PURPOSE, ELECTRICAL: W30U THROUGH W33U, W36U, W38U, W45U, W46U, W58U, W70U, W71U AND W73U

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for a series of twelve (12) electrical cable assemblies entitled as Cable Assemblies. The identification of cables are; W30U, W31U, W32U, W33U, W36U, W38U, W45U, W46U, W58U, W70U, W71U, W73U. The cables are used with the Lance Warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specifications For.
MIL-W-16878	- Wire, Electrical, Insulated, High Temperature.
MIL-C-27599	- Connector, Electrical, Miniature, Quick Disconnect (For Weapons System) Established Reliability.
MIL-M-60903	- Marking of Electrical Wires and Cables.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-331	- Fuze and Fuze Components, Environmental and Performance Tests For.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedures For.

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DRAWINGS

PICATINNY ARSENAL

8876233	- Cable Mandatory Process Information.
8876245	- Coaxial Cable & Connector Mandatory Process Information.
8880202	- Knob Assembly, Rotary Switch Number 2.
8880204	- W73U Cable Assembly, Option Selector Switch.
8880210	- W30U Cable, Control Unit to Station 100.
8880211	- W31U Cable, Warhead.
8880212	- W32U Cable, Fuze, Impact.
8880213	- W33U Cable, Fuze, Impact.
8880216	- W36U Cable, Fuze, Command.
8880218	- W38U Cable, PAL (Internal).
9236828	- Procedures for Lacing of Wiring Harness.
9236931	- W45U Cable, Impact Crystal.
9236932	- W46U Cable, Impact Crystal.
9236985	- W58U Cable, PAL (External).
9237149	- W70U Cable Assembly, Special Purpose, Electrical.
9237150	- W71U Cable Assembly, Special Purpose, Electrical.
9237196	- Mandatory Process Information, Coaxial Cable and Connector for W45U & W46U Assemblies.

PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1	- Quality Assurance Instructions for Reporting Data on Nuclear Weapons.
EL-8880204	- Equipment List.
EL-8880210	- Equipment List.
EL-8880211	- Equipment List.
EL-8880212	- Equipment List.
EL-8880213	- Equipment List.
EL-8880216	- Equipment List.
EL-8880218	- Equipment List.
EL-9236931	- Equipment List.
EL-9236932	- Equipment List.
EL-9236985	- Equipment List.
EL-9237149	- Equipment List.
EL-9237150	- Equipment List.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The cable assemblies shall comply with all requirements specified on the drawing (dwg) 8880204, 8880210, 8880211, 8880212, 8880213, 8880216, 8880218, 9236931, 9236932, 9236985, 9237149 and 9237150, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Coded serialization requirements (see 6.1.2). The contractor shall supply feedback data on punch cards for any item for which coding is required in section 3. The feedback data shall be in accordance with QAI-PA-1.

3.6 Mating Assembly. The cable assemblies shall be capable of being completely mated and unmated with their proper receptacles without the use of tools. This capability shall be verified as specified in 4.4.1.

3.7 Circuit resistance. The circuit resistance between the pins of the cable assemblies listed in Table I through XII shall be as specified when tested in accordance with 4.4.2.

Table I

W30U Cable, Control Unit to Sta. 100 (drawing 8880210
Resistance 0.030 Ohms (max.)

From 1P2	To 3P1	Test Code (See 3.14)	From 1P2	To 3P1	Test Code (See 3.14)
A	A	004001	K	E	004015
M	R	004002	S	S	004016
B	L	004003	L	H	004017
H	M	004004	Y	G	004018
C	K	004005	N	a	004019
R	J	004006	X	V	004020
D	D	004007	P	b	004021

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a	T	004008	U	F	004022
E	Z	004009	T	c	004023
V	Y	004010	Z	U	004024
F	B	004011	W	C	004025
c	P	004012	b	N	004026
G	K	004013	Shell	Shell	004027
J	W	004014			

Table II
W31U Cable, Warhead (dwg 8880211)
Resistance 0.100 Ohms (max)

From 1P12	To A7P2	Test Code (See 3.14)	From 1P12	To A7P2	Test Code (See 3.14)
A	A	004028	R	R	004042
T	T	004029	e	e	004043
F	F	004030	Z	Z	004044
D	D	004031	G	G	004045
Y	Y	004032	N	N	004046
c	c	004033	g	g	004047
h	h	004034	f	f	004048
J	J	004035	L	L	004049
M	M	004036	K	K	004050
X	X	004037	C	C	004051
E	E	004038	V	V	004052
U	U	004039	B	B	004053
P	P	004040	Shell	Shell	004054
H	H	004041	S	S	004239

Table III
W32U Cable, Fuze, Impact (dwg 8880212)
Resistance 0.150 ohms (max)

From Conductor	To Conductor	Test Code (see 3.14)	From Shell	To Shell	Test Code (see 3.14)
1P16	1P22	004055	1P16	1P22	004057
1P16	2P8	004056	1P16	2P8	004058

Table IV
W33U Cable, Fuze, Impact (dwg 8880213)
Resistance 0.150 ohms (max)

From Conductor	To Conductor	Test Code (see 3.14)	From Shell	To Shell	Test Code (see 3.14)
1P17	1P23	004059	1P17	1P23	004061
1P17	2P11	004060	1P17	2P11	004062

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Table V
W36U Cable, Fuze, Command (dwg 8880216)
Resistance 0.130 ohms (max)

From 1P1	To 1P28	To 1P29	Test Code (see 3.14)	From 1P1	To 1P28	To 1P29	Test Code (see 3.14)
				P		N	004077
A	T		004063	R	N		004078
				S	H		004079
B	E		004064	T		T	004080
				U	S		004081
C		J	004065	V		H	004082
				W		Y	004083
D	X		004066	X		X	004084
				Y		V	004085
				Z	b		004086
E	Y		004067	a		D	004087
F	V		004068	b		U	004088
G		b	004069	c		B	004089
				d	J		004090
H		M	004070	e		S	004091
				h	R		004092
J	D		004071	j		R	004093
				From	M	E	004094
K		a	004072	g		F	004095
L	a		004073	Shell	Shell		004096
M	U		004074	Shell		Shell	004097
N	B		004075				
f	F		004076				

Table VI
W38U Cable, PAL (Internal), (dwg 8880218)
Resistance 0.030 ohms (max)

From A2P1	To A2J2	Test Code (see 3.14)	From A2P1	To A2J2	Test Code (see 3.14)
A	A	004098	L	L	004107
B	B	004099	M	M	004108
C	C	004100	N	N	004109
D	D	004101	P	P	004110
E	E	004102	R	R	004111
F	F	004103	S	S	004112
G	G	004104	U	U	004113
H	H	004105	Shell	Shell	004114
K	K	004106	J	J	004235
			T	T	004236

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Table VII
W45U Cable, Impact Crystal (dwg 9236931)
Resistance 0.030 ohms (max)

From Conductor	To Conductor	Test Code (see 3.14)	From Shell	To Shell	Test Code (see 3.14)
1P25	2P12	004115	1P25	2P12	004116

Table VIII
W46U Cable, Impact Crystal (dwg 9236932)
Resistance 0.030 ohms (max)

From Conductor	To Conductor	Test Code (see 3.14)	From Shell	To Shell	Test Code (see 3.14)
1P24	2P9	004117	1P24	2P9	004118

Table IX
W58U Cable, PAL (External)(dwg 9236985)
Resistance 0.050 ohms (max)

From A2P2	To A2J3	Test Code (see 3.14)	From A2P2	To A2J3	Test Code (see 3.14)
A	A	004119	L	L	004128
B	B	004120	M	M	004129
C	C	004121	N	N	004130
D	D	004122	P	P	004131
E	E	004123	R	R	004132
F	F	004124	S	S	004133
G	G	004125	U	U	004134
H	H	004126	Shell	Shell	004135
K	K	004127	J	J	004237
			T	T	004238

Table X
W70U Cable Assembly Special Purpose, Electrical (dwg 9237149)
Resistance 0.075 ohms (max)

From P1	To P2	Test Code (see 3.14)	From P1	To P2	Test Code (see 3.14)
A	A	004136	M	M	004145
B	B	004137	P	P	004146
C	C	004138	R	R	004147
E	E	004139	S	S	004148
F	F	004140	T	T	004149
G	G	004141	U	U	004150

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J	J	004142	V	V	004151
K	K	004143	b	b	004152
L	L	004144	c	c	004153
			Shell	Shell	004233

Table XI

W71U Cable Assembly Special Purpose, Electrical (dwg 9237150)
Resistance 0.075 ohms (max)

From P1	To P2	Test Code (see 3.14)	From P1	To P2	Test Code (see 3.14)
L	30	004154	P	44	004163
A	31	004155	U	45	004164
B	33	004156	S	47	004165
C	34	004157	V	49	004166
E	36	004158	R	52	004167
F	37	004159	b	53	004168
G	38	004160	T	54	004169
J	40	004161	M	55	004170
K	42	004162	Shell	Shell	004234

Table XII
W73U Cable Assembly, Option Selector Switch

From P3	To P1	To P2	Resistance (ohms,max)	Test Code (see 3.14)
A		c	0.025	004171
B		S	0.025	004172
C	5		0.025	004173
D	17		0.025	004174
E		b	0.025	004175
F	30		0.025	004176
G		E	0.025	004177
H	3		0.025	004178
J	37		0.025	004179
K		V	0.025	004180
L		a	0.025	004181
M	16		0.025	004182
N	29		0.025	004183
P		M	0.025	004184
R	23		0.025	004185
S	6		0.025	004186
T		A	0.025	004187
U	18		0.025	004188
V	13		0.025	004189

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Table XII (con't)

From P3	To P1	To P2	Resistance (ohms, max)	Test Code (see 3.14)
W	7		0.025	004190
X	28		0.025	004191
Y	19		0.025	004192
Z	33		0.025	004193
a	31		0.025	004194
b	12		0.025	004195
c	14		0.025	004196
d	11		0.025	004197
e	8		0.025	004198
f	36		0.025	004199
g	1		0.025	004200
h	24		0.025	004201
j		P	0.025	004202
k		L	0.025	004203
m	20		0.025	004204
n	27		0.025	004205
p	9		0.025	004206
q	21		0.025	004207
r	34		0.025	004208
s		G	0.025	004209
t	26		0.025	004210
u		H	0.025	004211
v		J	0.025	004212
w	4		0.025	004213
x	15		0.025	004214
y		U	0.025	004215
z		T	0.025	004216
AA	25		0.025	004217
BB		Z	0.025	004218
CC	10		0.025	004219
DD	22		0.025	004220
EE		B	0.025	004221
FF	2		0.025	004222
GG		W	0.025	004223
HH		R	0.025	004224
JJ		Y	0.025	004225
KK		C	0.025	004226
LL		F	0.025	004227
MM		N	0.025	004228
NN		K	0.025	004229
Shell	Shell	D	0.025	004230
Shell		Shell	0.250	004231
			0.250	004232

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3.8 Insulation Resistance. The insulation resistance between each circuit and all other circuits and connector shell, shall be not less than 100 megohms when 500 ± 50 volts direct current (vdc) is applied. Testing shall be as specified in 4.4.3 with Tables XIII through XXIV. Unused terminals shall be connected together and tested as one terminal.

Table XIII
W30U Cable, Warhead (dwg. 8880210)

1P2 - A, B, C, D, E, F, G, H, J, K, L, M, N, P, R, S, T, U, V, W, X, Y, Z, a, b, c, Shell

Table XIV
W31U Cable, Warhead (dwg. 8880211)

1P12 - A, B, C, D, F, G, H, J, L, M, N, R, U, S, T, V, Y, P, Z, c, h, E, K, g, f, e, X, Shell

1P12 - Unused terminals: W, a, b, d, j

A7P2 - Unused terminals: W, j, d, a, b

Table XV
W32U Cable, Fuze, Impact (dwg. 8880212)

1P16 - Center Contact, Shell

Table XVI
W33U Cable, Fuze, Impact (dwg. 8880213)

1P17 - Center Contact, Shell

Table XVII
W36U Cable, Fuze, Command (dwg. 8880216)

1P1 - A, B, C, D, E, F, G, H, J, K, L, M, N, P, R, S, T, U, V, W, X, Y, Z, a, b, c, d, e, h, j, f, g, Shell
1P28 - M
1P28 - Unused terminals: A, C, G, K, L, P, W, Z, c
1P29 - Unused terminals: A, C, G, K, L, P, W, Z, c

Table XVIII
W38U Cable, PAL (Internal) (dwg. 8880218)

A2P1 - A, B, C, D, E, F, G, H, J, K, L, M, N, P, R, S, T, U, Shell

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Table XIX
W45U Cable, Impact Crystal (dwg 9236931)

1P25 - Center Contact, Shell

Table XX
W46U Cable, Impact Crystal (dwg 9236932)

1P24 - Center Contact, Shell

Table XXI
W58U Cable, PAL (External) (dwg 9236985)

A2P2 - A, B, C, D, E, F, G, H, J, K, L, M, N, P, R, S, T, U, Shell

Table XXII
W70U Cable Assembly Special Purpose, Electrical (dwg 9237149)

P1 - A, B, C, E, F, G, J, K, L, M, P, R, S, T, U, V, b, c
P1 - Unused terminals: D, H, N, W, X, Y, Z, a, and shell
P2 - Unused terminals: D, H, N, W, X, Y, Z, a

Table XXIII
W71U Cable Assembly Special Purpose, Electrical (dwg 9237150)

P3 - 55, 54, 53, 52, 49, 47, 45, 42, 40, 38, 37, 36, 34, 33, 31, 30
P3 - Unused terminals: 1 thru 29, 32, 35, 39, 41, 43, 46, 48, 50, 51, 56 thru 67, and Shell

Table XXIV
W73U Cable Assembly, Option Selector Switch (dwg 8880204)

P3 - A, B, C, D, E, F, G, H, J, K, L, M, N, P, R, S, T, U, V, W, X, Y, Z, a, b, c, d, e, f, g, h, i, j, k, m, n, p, q, r, s, t, u, v, w, x, y, z, AA, BB, CC, DD, EE, FF, GG, HH, JJ, KK, LL, MM, NN and shell
P3 - Unused terminals: PP
P2 - Unused terminals: X
P1 - Unused terminals: 32, 35

3.9 Capacitance. The capacitance between center conductor and the connector shell of the cable assemblies listed in Tables XXV through XXVIII shall be as specified when tested in accordance with 4.4.4.

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Table XXV
W32U Cable, Fuze Impact (dwg 8880212) Capacitance
255 ± 20 picofarads (pf) at 1000 ± 50 Hertz (Hz)

From Shell	Test Code (see 3.14)	To Center Conductor
1P16	006001	2P8
1P16	006002	1P22
1P16	006003	1P16

Table XXVI
W33U Cable, Fuze Impact (dwg 8880213) Capacitance
255 ± 20 picofarads (pf) at 1000 ± 50 Hertz (Hz).

From Shell	Test Code (see 3.14)	To Center Conductor
1P17	006004	2P11
1P17	006005	1P23
1P17	006006	1P17

Table XXVII
W45U Cable, Impact Crystal (dwg 9236931) Capacitance
23 ± 5 picofarads (pf) at 1000 ± 50 Hertz (Hz)

From Shell	Test Code (see 3.14)	To Center Conductor
1P25	006007	2P12
1P25	006008	1P25

Table XXVIII
W46U Cable, Impact Crystal (dwg 9236932) Capaitance
23 ± 5 picofarads (pf) at 1000 ± 50 Hertz (Hz)

From Shell	Test Code (see 3.14)	To Center Conductor
1P24	006009	2P9
1P24	006010	1P24

3.10 Temperature shock. Prior to and following the temperature shock test as specified in 4.4.5, the cable assembly shall comply with the requirements of 3.7, 3.8 and 3.9 within one (1) hour after having been returned to ambient conditions.

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3.11 Mechanical shock. The cable assembly shall be subjected to a half sine shock pulse of 25 ± 2.5 gravity units (g's) three times in each direction along each of the three mutually perpendicular axes (18 shock total). The duration of each shock shall last for 11 ± 2 milliseconds (ms) as measured between points which are 10 percent of the peak level. The time to peak shall be 5 ± 1 ms. The cable assembly shall be tested as specified in 4.4.6. Prior to and upon completion of this test, the cable assembly shall comply with the requirements of 3.7, 3.8 and 3.9 and shall show no evidence of connector damage.

3.12 Transportation vibration. The cable assembly shall be subjected in each of its three mutually perpendicular axes to the vibration levels for the lengths of times shown in Table XXIX at each of the specified temperatures. After completion of the vibration tests, and upon returning to room ambient temperature without forced cooling or heating, the cable assembly shall comply to the requirements of 3.7, 3.8 and 3.9 and shall show no evidence of connector damage. The cable assembly shall be tested as specified 4.4.7.

Table XXIX

Frequency (Hz)*	Time Per Sweep (minutes)	Time (Minutes) Temperature (°F) -40°F	+140°F	Double Amplitude da or g Level
10-50-10	2.0	16	16	0.04 inch da
50-115-50	2.5	10	10	5.0 g
115-500-115	2.0	10	10	10.0 g
	TOTAL	36	36	

*Hertz

3.13 Handling.

3.13.1 Bend (W30U, W31U, W36U, W38U, W58U, Cables). The cable assembly shall show no evidence of distortion, cracks, bend separation, or any other damage when subjected to flexing around mandrels specified in 4.4.8.1. Wrinkling of the cable jacket during flexure is acceptable. Cable shall bend freely around specified mandrel. Space between cable and mandrel shall not exceed 0.06 inches.

3.13.2 Twist-Degrees (W30U, W31U, W36U, W38U, W58U, Cables). The cable assembly shall show no evidence of cracks, mold bond separation, or any other damage when twisted as specified in 4.4.8.2. Cable assembly shall comply to the requirements of 3.7 and 3.8 upon completion of tests. Distortion of cable jacket during twist is acceptable.

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3.13.3 Twist-Torque (W32U, W33U, W45U, W46U, W73U, Cables). The cable assembly shall show no evidence of cracks, mold bond separation, slippage of cable in connector, or any other damage when twisted as specified in 4.4.8.3. Cable assembly shall comply to the requirements of 3.7, 3.8 and 3.9 upon completion of tests.

3.14 Test identification number. The contractor shall provide variables reading on punch cards using the six digit number designated in the applicable paragraph in Section 3. Punch cards shall be prepared and mailed in accordance with QAI-PA-1.

3.15 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.16 Workmanship.

3.16.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.16.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.16.3 Burrs. Parts shall be free from burrs.

3.16.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.16.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.16.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. The contractor shall submit first article samples (see 4.2.1) at the start of all new production contracts (see 6.5).

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4.2.1 Submission. The contractor shall submit 2 of W30U, 2 of W31U, 2 of W38U, 2 of either W32U or W33U, and 2 of either W45U or W46U, 2 of W36U, 1 each of W70U and W71U, 2 of W58U and 4 of W73U cables consecutively produced assemblies, which have been produced by the production process which the contractor will use in fulfulling the contract, to a Government approved facility designated (see 6.3) by the procuring activity. The first article quantity shall be submitted prior to initiation of production and whenever a change occurs in manufacturing process, material used, drawing or specification, such as to significantly affect product uniformity as determined by the procuring agency. The first article samples, after completion of the tests, shall be forwarded to Picatinny Arsenal for disposition (see 6.4).

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table

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Table XXX
First article inspection and tests

Inspection or Test	Cable assemblies W31U, W32U and W38U, W45U	W33U or W36U, and W46U,	W70U, W30U and W71U,	and W58U,	W73U,	Dest,	Non-Dest
1. Classification of characteristics (see 4.3.2.1)	X	X	X	X	X		X
2. Mating assembly (see 4.4)	X	X	X	X	X		X
3. Circuit resistance (see 4.4.2)	X	X	X	X	X		X
4. Insulation resistance (see 4.4.3)	X	X	X	X	X		X
5. Capacitance (see 4.4.4)		X					X
6* Temperature shock (see 4.4.5)	X	X	X	X	X	X	
7* Mechanical shock (see 4.4.6)	X	X	X	X	X	X	
8* Trans. vibrat @-40°F (see 4.4.7.1)	X	X	X	X	X	X	
9* Trans. vibrat @140°F (see 4.4.7.2)	X	X	X	X	X	X	
10. Handling (see 4.4.8)	X	X	X	X	X		X

* After tests 1 through 5 are completed, the cables shall be subjected to requirements of 3.7, 3.8 and 3.9 only once after test 10 is completed.

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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Inspection lot formation. All items in each inspection lot shall be from one production line, one shift, one batch of products, one manufacturer or raw material, one manufacturer of components, one production method, and one manufacturing operation. The entire lot shall be submitted as a unit for inspection.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table XXXI), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristics inspections are performed.

Table XXXI

Major Characteristics		Minor Characteristics	
Lot Size	No. of Sample (s)	Lot Size	No. of Sample(s)
1-25	Same as Lot Size	1-25	Same as Lot Size
26-30	25	26-30	25
31-35	27	31-35	26
36-40	30	36-40	27
41-45	32	41-45	28
46-50	34	46-50	29
51-55	35	51-55	30
56-60	37	56-60	31
61-65	38	61-65	32
66-70	39	66-70	33
71-75	40	71-75	34
76-85	42	76-85	35
86-95	43	86-95	36
96-110	44	96-110	37
111-130	46	111-130	38
131-156	48	131-156	39
157-190	49	157-190	40

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Table XXXI (con't)

191-240	51	191-240	41
241-340	53	241-340	42
341-440	54	341-520	43
441-540	55	521-1120	44
541-840	56	1121 and over	45
841-1500	57		
1501 and over	59		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Cable assemblies - before molding (all applicable dwgs see 3.2).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Any component missing, damaged loose or improper	Visual
	102. Improper connector	Visual
	103. Improper soldering (see 3.15)	Visual
	104. Improper pre-pot cleaning (see 2.3 of dwg 8876233)	Visual/Gage
	105. Improper attachment of shielding braid (see 2.4 of dwg 8876233]	Visual
	106. Presence of chips and foreign matter	Visual
	107. Evidence of poor workmanship (see 3.16)	Visual
	108. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

4.3.2.1.2 Cable assemblies - after molding (all applicable dwgs (see 3.2).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Incorrect length	Gage
	102. Incorrect connector orientation	Visual
	103. Incorrect physical configuration	Visual
	104. Lead wire not completely covered by molded rubber	Visual

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	105.	Any component missing, damaged loose or improper.....Visual
	106.	Incorrect mating terminal used in electrical tests (see 2.8.2 of dwg 8876233).....Visual
	107.	Poor bond between molded rubber and tubing.....Visual/Manual
	108.	Poor bond between molded rubber and connector shell.....Visual
	109.	Objective evidence relating to material missing or improper (see 3.1).....Visual
Minor:		
	201.	Marking incorrect, missing or illegible.....Visual
	202.	Evidence of surface voids not in accordance with 2.11.4 of dwg 8876233.....Visual
	203.	Evidence of poor workmanship (see 3.16).....Visual

4.3.2.1.3 Cable assembly, option selector switch, W73U (see dwg 8880204), before molding.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101.	Incorrect length of shield braid, 3 places.....Gage
	102.	Incorrect washer for potting shell.....Visual
	103.	Incorrect minimum length of shrinkable tubing P1 only, 35 places.....Gage
	104.	Improper soldering leads (P1 only) to adapter pin, 35 places (see 3.15)...Visual
	105.	Improper solder P1 adapter pins to P1 connector, 35 places (see 3.15).....Visual
	106.	Improper attachment and wrapping of RF shielding.....Visual/Gage
	107.	Improper connectors (3).....Visual
	108.	Presence of chips and foreign matter...Visual
	109.	Improper lacing (see 9236828).....Visual
	110.	Evidence of poor workmanship (see 3.16).....Visual
	111.	Objective evidence relating to material missing or improper (see 3.1).....Visual
Minor:	None defined.	

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4.3.2.1.4 Cable assembly, option selector switch, W73U (see dwg 8880204), after molding.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Incorrect connector orientation, 3 places	Visual
	102. Incorrect connector location, 3 places	Gage
	103. Incorrect location of connectors major keyways, 3 places	Gage
	104. Excessive voids or bubbles	Gage
	105. Excessive voids exposing shielding	Visual
	106. Incorrect physical configuration	Visual
	107. Poor bond between molded rubber and potting shell, 3 places	Visual
	108. Poor bond between molded rubber and tubing	Visual/Manual
	109. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Improper or missing encapsulation (see 8880202)	Visual
	202. Evidence of poor workmanship (see 3.16)	Visual

4.3.2.1.5 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Preservation packing and packaging not in accordance with MIL-STD-1169	Visual
	102. Marking not in accordance with MIL-STD-1169	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.16)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

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	Test	Test classification
a.	Mating assembly	Major
b.	Circuit resistance	Major
c.	Insulation resistance	Major
d.	Capacitance	Major
e.	Transportation vibration at 165°F	Major
f.	Transportation vibration at -65°F	Major

Of these tests, e and f are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Mating assembly circuit resistance, insulation resistance W30U, W31U, W32U, W33U, W36U, W38U, W45U, W46U, W58U, W70U, W71U, W73U. Each cable assembly shall be tested in accordance with 4.4.1, 4.4.2 and 4.4.3. If any item fails to comply with 3.7, 3.8, and 3.9, the item shall be rejected and removed from the lot. Variables data on punch cards are not required for these tests.

4.3.3.2 Capacitance W32U, W33U, W45U, W46U. Two of each kind of cable assembly shall be selected at random from each inspection lot and shall be tested in accordance with 4.4.4. If any sample cable assembly fails to comply with 3.9, the lot shall be rejected.

4.3.3.3 Transportation vibration W30U, W31U, W32U, W33U, W36U, W38U, W58U, W73U. Two cables shall be selected at random from each group listed below in Table XXXII and be tested in accordance with 4.4.7. If any sample cable fails to comply with 3.12, the lot shall be rejected.

Table XXXII

Group #1	Group #2	Group #3	Group #4	Group #5
W31U	W32U	W36U	W58U	W73U
W38U	W33U		W30U	

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8880204, 8880210, 8880211, 8880212, 8880213, 8880216, 8880218, 9236931, 9236932, 9236985, 9237149, 9237150). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

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a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2)

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Mating assembly. The cable assembly shall mate to connectors equivalent to the terminations they are to be used on so that an audible click is evident, and that visual observation of the locking pins is confirmed, when proper coupling is accomplished.

4.4.2 Circuit resistance. The resistance between the pins listed in 3.7 shall be measured with a Government approved electrical resistance tester. The resistance shall not exceed the values specified in 3.7. The current applied shall be limited to 100 milliamperes (ma) maximum.

4.4.3 Insulation resistance. Insulation resistance shall be measured by means of Government approved instrument capable of measuring megohms. Any circuit exhibiting an insulation resistance less than 100 megohms shall be accepted if the observed value increased to 100 megohms minimum after the specified voltage has been applied for 60 seconds max. The current applied shall be limited to 5 ma maximum.

4.4.4 Capacitance (W32U, W33U, W45U, W46U). The capacitance between center pin and connector shell shall be measured with a Government approved electrical instrument, with a suitable coaxial connector, capable of measuring picofarads. The capacitance shall be within values specified in 3.8.

4.4.5 Temperature shock. The cable assembly with its connectors covered by protective caps shall be placed in test chambers, one capable of maintaining 140 $\pm$ 5°F and the other of maintaining -40° $\pm$ 5°F. The cable assembly shall be subjected to temperature shock as specified in MIL-E-5272C procedure I. Upon completion of the test, the cable assembly shall comply with the requirements specified in 3.10.

4.4.6 Mechanical shock. The cable assembly shall be mounted on a suitable shock producing device. This will be accomplished in the following manner: The cables shall be mounted to an approved fixture by attaching each cable connector to a mating connector mounted on the approved fixture.

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The length of cable between connectors shall not be attached to either the fixture, the shock machine, or any stationary support, but shall hang unsupported. Spacing of the connectors on the fixture need not simulate final cable application position. The cable assembly shall be subjected to the requirements specified in 3.10.

4.4.7 Transportation vibration.

4.4.7.1 At low temperature (-40°F). The cable assembly shall be rigidly mounted to an approved fixture capable of being readily positioned in each of the three mutually perpendicular axes. The cables shall be mounted to the approved fixture by attaching each cable connector to a mating connector mounted on the fixture. Spacing the connectors on the fixture need not simulate final cable application position. The fixture shall then be mounted on a suitable vibration producing apparatus. The cable assembly shall be vibrated at the frequencies, amplitudes and for the times specified in Table XXIX.

4.4.7.2 At high temperature (+140°F). The cable assembly shall be rigidly mounted to an approved fixture capable of being readily positioned in each of the three mutually perpendicular axes. The cables shall be mounted to the approved fixture by attaching each cable connector to a mating connector mounted on the fixture. Spacing of the connectors on the fixture need not simulate final cable application position. The fixture shall then be mounted on a suitable vibration producing apparatus. The cable assembly shall then be vibrated at the frequencies, amplitudes and for the times specified in Table XXIX.

4.4.8 Handling.

4.4.8.1 Bend (W30U, W31U, W38U, W58U Cables). Starting with jacket end and molding touching mandrel, bend the cable between 150° and 180° per portion, around mandrel. Remove this portion from the mandrel and repeat bends of 150° (min) with consecutive portions until entire unmolded portion of cable has been bent. Do not bend more than 180° any length of cable at one time.

Cable Assembly	Mandrel Dia (+0.06)
W36U P1P Branch	1.5
1P29 Branch	1.5
1P28 Branch	1.5
W31U, W38U, W58U	1.5
W30U	2.0

4.4.8.2 Twist (degrees) (W30U, W31U, W36U, W38U, W58U,). Apply a torque to the cable using simulated mating connectors, or with the molded joint clamped, so as to obtain a twist of 90° + 5° per foot of cable length. This twist shall be applied first in a clockwise direction and then in a counterclockwise direction after the cable has returned to its initial at rest position.

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All nonmolded portions of cable shall be tested.

4.4.8.3 Twist (torque) (W32U, W33U, W45U, W46U, W73U, Cables). Apply a torque of 3+1 pound inches to cable, using mating connectors which are rigidly secured or with molded joints clamped. This torque shall be applied first in a clockwise direction and then in a counter-clockwise direction after the cable has been returned to its initial at rest position (1 cycle). All branches of cable shall be tested. This test shall be performed in 3 cycles.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The cable assemblies shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169 .

5.1.2 Level C. The cable assemblies shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169 .

5.2 Packing.

5.2.1 Levels A and C. The cable assemblies shall be packed in accordance with the applicable instructions contained in MIL-STD-1169 .

5.3 Marking. Marking shall be in accordance with MIL-STD-1169

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 3.7, 3.9, and 4.3.4.1). Data specified in 3.7, 3.9 and 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

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6.5 Special first article conditions. First article samples will be required unless either of the following conditions exist:

a. An uninterrupted production of the identical item by the same manufacturer.

b. Ninety days have not elapsed since the identical item was produced by the same manufacturer presently under contract. Testing under Research and Development contracts shall not be caused for waiver or suspension of first article testing.

6.6 Supersession data. This specification supersedes the requirements of Purchase description PA-PD-2645A dated 20 February 1973, with Amendment 3, dated 25 July 1974.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A216

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